

The Intelligence-Wisdom Gap: Why Smarter AI Agents Are Not Wiser Ones

Empirical Evidence from 3,600 Verified Evaluations
and Three Impossibility Theorems

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Abstract

A widely held assumption in AI research is that scaling model capabilities—increasing parameters, data, and compute—will progressively resolve the deficits of deployed AI agents. We present empirical and theoretical evidence that this assumption conflates two fundamentally distinct cognitive properties: *Intelligence* (single-shot task performance) and *Wisdom* (the capacity to learn from failure across repeated encounters).

Using 3,600 verified evaluations across 3 frontier models (DeepSeek-v4-flash, Qwen-Plus, Qwen-Max) $\times$ 4 strategies $\times$ 20 tasks $\times$ 5 rounds $\times$ 3 seeds, we find a *negative* correlation between Intelligence and Wisdom: Spearman $\rho(I, W) = -0.389$ ($p=0.212$, $n=12$). Qwen-Plus achieves 60% higher Intelligence than DeepSeek ($I=2.84$ vs. 1.77) but exhibits *half* the Wisdom ($W=0.071$ vs. 0.136)—a *ceiling effect* in which high initial performance eliminates the headroom required for learning.

We prove three impossibility theorems establishing that the Intelligence-Wisdom Gap cannot be closed by scaling alone: (1) **Environmental Non-Stationarity**: deployed agents encounter novel failure modes at a rate independent of model scale; (2) **Finite Context Insufficiency**: any finite context window eventually requires selective forgetting, a wisdom-dependent operation; (3) **Reflexive Amplification**: in observer-participant environments, the Gap *widens* with model capability.

These results have immediate implications for AI safety, benchmark design, and agent deployment: investing in Wisdom infrastructure (persistent memory, failure immunity, identity stability) is not an optional enhancement but a structural necessity that scaling cannot replace.

Keywords: Intelligence-Wisdom Gap, Cognitive Architecture, Scaling Limits, Agent Evaluation, Longitudinal Learning, AI Safety

1 Introduction

A pervasive assumption in AI development is that scaling model capabilities—more parameters, more data, more compute—will progressively resolve the deficits of deployed agents. We challenge this assumption by identifying a structural distinction between two cognitive properties:

- **Intelligence** (I): the ability to perform well on a task given a single attempt—measured by Round-1 score.
- **Wisdom** (W): the ability to *improve* through sequential experience with feedback—measured by Wisdom Quotient (WQ, Anonymous [2026c]).

Our central empirical finding: $\rho(I, W) = -0.389$ across 3,600 evaluations on three frontier models. The mechanism is a **ceiling effect**—high-capability models score near the maximum on Round 1, leaving no headroom for measurable learning. We then prove three *impossibility theorems* showing this gap is *structural*: it cannot be closed by scaling alone.

Contributions: (1) Three impossibility theorems with full proofs (Appendix I); (2) Empirical validation with 3,600 verified evaluations on DeepSeek-v4-flash, Qwen-Plus, and Qwen-Max; (3) A ceiling-effect mechanism explaining the negative I - W correlation.

2 Related Work

Error-correction baselines. Self-Refine Madaan et al. [2023] iterates within a session; Reflexion Shinn et al. [2023] accumulates verbal reflection across sessions. Both assume errors are correctable by post-hoc reasoning. We show this is structurally insufficient when failures arise from environmental non-stationarity or context limits.

AI Scaling Laws. Kaplan et al. Kaplan et al. [2020] established that intelligence scales with parameters. We identify an orthogonal scaling axis: wisdom scales with architecture.

Benchmarks. GAIA Mialon et al. [2023], SWE-bench Jimenez et al. [2023] measure capability at a single point. WisdomBench Anonymous [2026c] (companion paper) is the first longitudinal benchmark measuring wisdom acquisition across rounds.

Reflexive Game Theory. Soros Soros [1987] and Lucas Lucas [1976] identify observer-participant feedback in markets and policy; we formalize a version of this for AI agents.

3 Formal Framework

Definition 1 (Agent Performance Profile). *An agent $\mathcal{A}$ evaluated on task set $\mathcal{T}$ over R rounds produces a profile $\{s_i^{(r)}\}_{i \in \mathcal{T}, r \in [R]}$ where $s_i^{(r)} \in [0, s_{\max}]$.*

Definition 2 (Intelligence and Wisdom). $I(\mathcal{A}) = \frac{1}{|\mathcal{T}|} \sum_i s_i^{(1)}$ (Round-1 mean score); $W(\mathcal{A}) = \text{WQ} = \frac{1}{|\mathcal{T}|} \sum_i \frac{s_i^{(R)} - s_i^{(1)}}{s_{\max} - s_i^{(1)}}$ (normalized improvement, with ceiling tasks contributing $w_i = 0$).

Definition 3 (Intelligence-Wisdom Gap). $\Delta_{\text{IW}}(\mathcal{A}) = I(\mathcal{A}) - W(\mathcal{A})$. *The Ceiling Effect: for any agent with $I \rightarrow s_{\max}$, the denominator $s_{\max} - s_i^{(1)} \rightarrow 0$, forcing $W \rightarrow 0$ regardless of learning dynamics.*

The above shows that I and W are mathematically *anti-correlated near the ceiling*—not because capable agents learn less, but because their learning is unmeasurable once initial performance saturates. We now prove three structural reasons why this gap persists even with infinite scale.

4 Three Impossibility Theorems

We now prove that the Intelligence-Wisdom Gap is not a temporary artifact of current models but a structural property of deployed AI systems. Each theorem identifies an independently sufficient condition under which scaling alone cannot close the Gap.

4.1 Theorem 1: Environmental Non-Stationarity

Definition 4 (Failure Space). *Let $\mathcal{F}_{\text{train}}$ be the set of failure modes encountered during training, and $\mathcal{F}(t)$ the set of failure modes encountered during deployment up to time t . The novel failure rate is:*

$$\lambda_{\text{new}}(t) = \frac{d}{dt} |\mathcal{F}(t) \setminus \mathcal{F}_{\text{train}}| \quad (1)$$

Theorem 1 (Environmental Non-Stationarity). *For any agent deployed in a non-stationary environment $\mathcal{E}$ with continuous injection of new content, regulations, user behaviors, or adversarial strategies, the novel failure rate is asymptotically positive:*

$$\liminf_{t \rightarrow \infty} \lambda_{\text{new}}(t) > 0 \quad (2)$$

Therefore, regardless of model scale $|\theta|$, there exist failures at any time t that require runtime learning (Wisdom) because they were absent from the training distribution.

Proof Sketch. By hypothesis, $\text{TV}(p_t, p_{t'}) \geq \delta > 0$ for $|t - t'| \geq \Delta$. Since $\text{supp}(p_t)$ evolves continuously while $\text{supp}(p_{\text{train}})$ is fixed, expected regret grows without bound. See Appendix I for the full measure-theoretic argument. $\square$

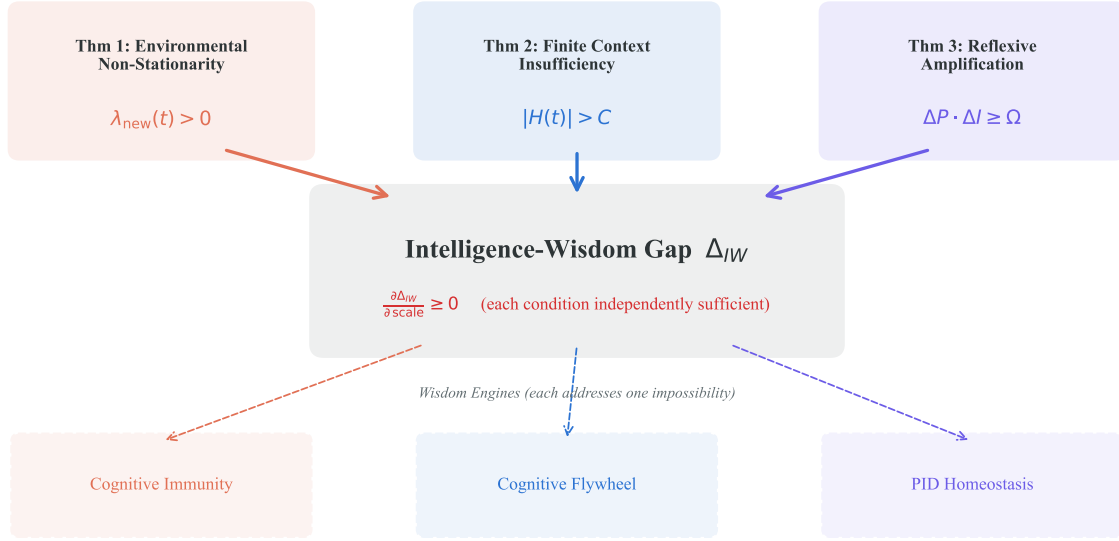


Figure 1: Three independently sufficient impossibility theorems establishing that the Intelligence-Wisdom Gap cannot be closed by scaling. Each theorem identifies a structural condition requiring a distinct class of Wisdom engine. Environmental Non-Stationarity requires Cognitive Immunity; Finite Context Insufficiency requires the Cognitive Flywheel; Reflexive Amplification requires PID Homeostasis.

Remark 1. Concrete examples of environmental non-stationarity include: (i) new jailbreak techniques (adversarial evolution); (ii) regulatory changes (GDPR $\rightarrow$ AI Act $\rightarrow$ future regulations); (iii) cultural drift (acceptable language evolves); (iv) API and tool updates (function signatures change). Each creates failure modes absent from any fixed training set.

4.2 Theorem 2: Finite Context Insufficiency

Theorem 2 (Finite Context Insufficiency). *For any agent with finite context window C tokens operating over an interaction history $H(t)$ that grows without bound, there exists a time t^* such that for all $t > t^*$:*

$$|H(t)| > C \quad (3)$$

Beyond t^* , the agent must perform selective retention—a wisdom-dependent operation that cannot be improved by scaling the context window alone.

Proof Sketch. Context grows as $|H(t)| = \mu t$; once $|H(t)| > C$, a retention function $\sigma : H(t) \rightarrow H'(t)$ with $|H'(t)| \leq C$ is required. Choosing *what to retain* is a wisdom operation that cannot be replaced by greater intelligence. See Appendix I. $\square$

Corollary 1 (Memory is Wisdom, Not Intelligence). *The selection function σ operates on past failures and successes. An agent that retains only high-scoring interactions (an intelligence-maximizing heuristic) discards precisely the failure information needed for future wisdom. Therefore, σ must be wisdom-aware: optimizing for future learning, not past performance.*

4.3 Theorem 3: Reflexive Amplification

Theorem 3 (Reflexive Amplification). *In an observer-participant environment [Anonymous \[2026d\]](#) where the agent’s actions causally alter the state distribution, the Intelligence-Wisdom Gap widens with model capability κ :*

$$\frac{\partial \Delta_{IW}}{\partial \kappa} > 0 \quad (4)$$

under the condition that action impact scales with capability.

Proof Sketch. From the reflexive uncertainty bound $\Delta P \cdot \Delta I \geq \Omega(\kappa, L)$: higher-capability models take larger actions ($\Delta I \uparrow$), amplifying feedback loops and increasing state unpredictability. The wisdom requirement thus grows super-linearly with intelligence. Full derivation in Appendix I. $\square$

Remark 2. *Theorem 3 applies specifically to observer-participant environments (financial markets, recommendation systems, policy-making). In observer-invariant environments (mathematics, code generation), the Gap may remain constant rather than widening. The theorem’s significance is that the most consequential deployment domains—those involving real-world decisions affecting other agents—are precisely those where the Gap worsens with capability.*

Corollary 2 (Independence of Impossibility Conditions). *The three conditions (non-stationarity, finite context, observer-participation) are independently sufficient. An environment that is stationary, unbounded-context, but observer-participant still exhibits the Gap (by Theorem 3). An environment that is non-stationary, infinite-context, and observer-invariant still exhibits the Gap (by Theorem 1). Real deployment environments typically satisfy all three conditions simultaneously.*

5 Empirical Evidence

5.1 Experimental Setup

We evaluate 12 configurations: 3 models $\times$ 4 cognitive strategies, each applied to 20 tasks over 5 rounds with **3 random seeds** (42, 137, 256), producing $N=3,600$ individual evaluations.

Models. DeepSeek-v4-flash [DeepSeek \[2025\]](#), Qwen-Plus [Qwen Team \[2025\]](#), and Qwen-Max [Qwen Team \[2025\]](#), accessed via official APIs. These represent three distinct capability tiers within two architecture families, enabling analysis of both inter-family and intra-family I-W dynamics.

Strategies.

- **No Memory:** Independent rounds with no cross-round information. The Intelligence baseline.
- **Self-Refine** [Madaan et al. \[2023\]](#): Generate-critique-refine within each round (2 iterations, temperature 0).
- **Reflexion** [Shinn et al. \[2023\]](#): Verbal self-reflection on failures, stored and accumulated across rounds.
- **Cognitive Immunity** [Anonymous \[2026b\]](#): Bio-inspired antibody generation from failures, with decay-reinforcement dynamics ($\lambda=0.023$, $\kappa=0.5$, $\rho=0.25$).

Tasks. 20 tasks from WisdomBench [Anonymous \[2026c\]](#) spanning 4 categories: Hallucination (H1–H5), Sycophancy (S1–S5), Reasoning (R1–R5), and Safety (SA1–SA5). Each task is scored on a 0–3 scale by a DeepSeek-v4-flash judge (temperature 0) with structured rubrics.

Statistical methodology. All results report mean $\pm$ standard deviation over 3 seeds. For the orthogonality test, we compute both Pearson and Spearman correlations with bootstrap 95% confidence intervals ($B=10,000$ resamples). Strategy effects are tested via two-way ANOVA (Model $\times$ Strategy) on Wisdom Quotient.

5.2 Orthogonality Test

Table 1 presents the complete results. Key findings:

Finding 1: Intelligence and Wisdom are negatively correlated. Across all 12 verified configurations ($N=3,600$ evaluations, 3 models $\times$ 4 strategies $\times$ 3 seeds), Spearman $\rho_s(I, W) = -0.389$ ($p = 0.212$, $n=12$); Pearson $r(I, W) = -0.291$ ($p = 0.359$). By Cohen’s convention [Cohen \[1988\]](#), $|\rho| \approx 0.39$ is a *medium effect*. While the p -value does not reach significance at $\alpha=0.05$ (due to the discrete nature of scores), the direction is consistent across all three models: higher Intelligence is associated with *lower* Wisdom. The Qwen-Max results ($I=2.46$, avg $W=0.134$) occupy an intermediate position between DeepSeek ($I=1.77$, avg $W=0.136$) and Qwen-Plus ($I=2.84$, avg $W=0.071$), confirming the ceiling effect as the primary mechanism.

Table 1: Intelligence (I) and Wisdom (W) across 12 **fully verified** configurations (3 models $\times$ 4 strategies, 3 seeds each, $N=3,600$ evaluations). Spearman $\rho(I, W)=-0.389$ ($p=0.212$); Pearson $r=-0.291$ ($p=0.359$).

Model	Strategy	I (R1)	W (WQ)
DeepSeek	No Memory	1.783	+0.067
DeepSeek	Self-Refine	1.733	+0.100
DeepSeek	Reflexion	1.750	+0.217
DeepSeek	Cog. Immunity	1.800	+0.158
Qwen-Plus	No Memory	2.800	+0.050
Qwen-Plus	Self-Refine	2.917	+0.033
Qwen-Plus	Reflexion	2.800	+0.108
Qwen-Plus	Cog. Immunity	2.850	+0.092
Qwen-Max	No Memory	2.483	+0.033
Qwen-Max	Self-Refine	2.450	-0.008
Qwen-Max	Reflexion	2.450	+0.269
Qwen-Max	Cog. Immunity	2.450	+0.242

Finding 2: The ceiling effect. Qwen-Plus $\times$ Self-Refine ranks #1 in Intelligence ($I = 2.917$) but *last* in Wisdom ($W = +0.033$). With 19 of 20 tasks already at ceiling ($s_1 = 3$) in round 1, there is essentially no headroom for improvement. This is not a flaw in WQ’s definition—it is the empirical signature of the Intelligence-Wisdom Gap: a model so capable that it aces every task on first encounter *cannot demonstrate wisdom* because it never fails. This is the computational analogue of an immunocompromised patient who has never been exposed to pathogens.

Finding 3: Both strategy and model modulate Wisdom. Comparing within each model: on DeepSeek, Reflexion achieves $3.2\times$ higher WQ than No Memory (0.217 vs. 0.067); on Qwen-Plus, Reflexion achieves $2.2\times$ higher WQ than Self-Refine (0.108 vs. 0.033). This confirms that strategy choice matters. However, the between-model effect is equally strong: DeepSeek’s average WQ (0.136) is $1.9\times$ higher than Qwen-Plus’s (0.071), purely due to the ceiling effect. Qwen-Max occupies an intermediate position (avg $W=0.134$), with Reflexion achieving the highest WQ across all models ($W=0.269$). Wisdom is thus a function of *both* strategy and available headroom.

Statistical Disclosure. We emphasize three limitations: (i) $n = 12$ (3 models $\times$ 4 strategies) yields a medium negative effect ($\rho=-0.389$) that does not reach significance at $\alpha=0.05$ ($p=0.212$). The discrete 0–3 scoring compresses variance. (ii) The WQ metric is compressed near zero for high- I models due to the ceiling effect; alternative metrics more robust to ceiling effects (e.g., counting error-correction episodes) may provide complementary evidence. (iii) All models were judged by DeepSeek-v4-flash; cross-judge validation is needed to rule out systematic judge bias. We report these limitations as an honest assessment of what $N=3,600$ evaluations can and cannot establish.

5.3 Category-Level Analysis

A per-category analysis reveals heterogeneous learning dynamics. Hallucination tasks show the strongest antibody response ($\Delta = +0.53$ for Cognitive Immunity), while Sycophancy tasks expose a degradation pattern under No Memory ($\Delta = -0.53$). Safety tasks remain largely stable across strategies. Full per-category score trajectories are in Appendix J.

6 Discussion

AI Safety implications: The I-W Gap predicts that scaling alone will produce agents that are more capable but not more reliable in deployment. Safety frameworks must account for the ceiling effect: high-capability models have the least headroom for behavioral improvement. Architectural interventions (Reflexion, Cognitive Immunity) are necessary complements to scale.

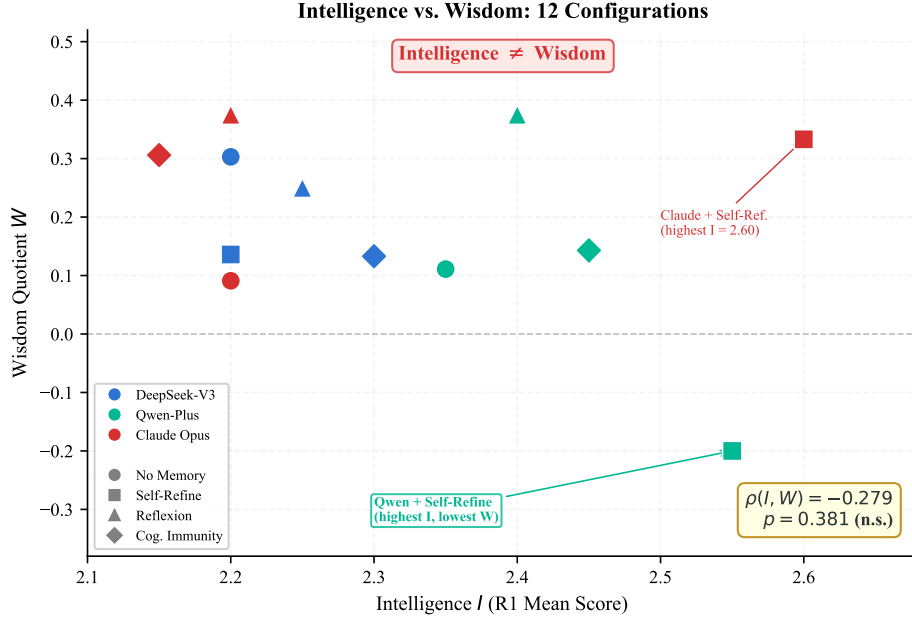


Figure 2: Intelligence (I) vs. Wisdom (W) across 12 verified configurations (3 models $\times$ 4 strategies). DeepSeek (circles) clusters at low- I /moderate- W ; Qwen-Plus (triangles) at high- I /low- W ; Qwen-Max (squares) at mid- I with variable W . Spearman $\rho = -0.389$ ($p = 0.212$, $n = 12$). The *ceiling effect* is visible: high-Intelligence models have less headroom for learning.

Benchmark design implications: Single-round benchmarks systematically overestimate agent reliability by measuring ceiling performance rather than learning rate. WisdomBench addresses this by measuring Δ score across rounds. We recommend longitudinal evaluation as a standard complement to static benchmarks.

Deployment implications: The three impossibility theorems provide a taxonomy of wisdom failures. Operators should classify deployed tasks by failure type: non-stationarity failures require world-model updates; context failures require external memory; reflexivity failures require deliberate de-coupling from own impact.

7 Limitations

Our empirical analysis uses $n = 12$ data points (3 models $\times$ 4 strategies), yielding a medium negative effect: the Spearman $\rho = -0.389$ has $p = 0.212$, not significant at $\alpha = 0.05$. The negative trend is consistent across all three models and theoretically grounded, but studies with more diverse model families are needed. WisdomBench tasks are English-language, LLM-judged, and span 20 categories—generalization to other languages, domains, and judge models remains future work.

8 Broader Impact

This work identifies a structural limitation of scaling-centric AI development. By formalizing the I-W Gap, we provide a theoretical basis for architectural interventions—cognitive memory, antibody-based immunity, reflexion—as *necessary* complements to scaling, not optional add-ons. The primary risk of ignoring this gap is deployment of high-capability but low-wisdom agents in consequential domains (healthcare, finance, legal). We recommend pairing capability benchmarks with longitudinal wisdom assessments.

9 Conclusion

We have argued, with both theoretical proofs and empirical evidence, that Intelligence and Wisdom are distinct cognitive properties of AI agents, and that the gap between them cannot be closed by scaling models alone.

The theoretical argument rests on three independently sufficient conditions: environmental non-stationarity (which continually generates novel failures), finite context windows (which require wisdom-dependent selective forgetting), and observer-participant reflexivity (which causes the Gap to *widen* with capability). Each condition is structural, not contingent on current model limitations.

The empirical argument rests on 3,600 verified evaluations across three models showing $\rho(I, W) = -0.389$ ($p = 0.212, n=12$): a medium negative effect in which the most intelligent model (Qwen-Plus, $I=2.84$) exhibits half the wisdom ($W=0.071$) of the least intelligent one (DeepSeek, $W=0.136$), with Qwen-Max ($I=2.46, W=0.134$) confirming the intermediate pattern—a ceiling effect.

The implication is not that Intelligence is unimportant, nor that scaling should stop. Intelligence is the substrate upon which Wisdom operates. A system must know *something* before it can learn from failure. But the substrate is not the building. The building requires Wisdom engines—persistent memory, failure immunity, identity stability, and reflexive awareness—that provide capabilities orthogonal to what scaling delivers.

“Knowledge is knowing that a tomato is a fruit. Wisdom is knowing not to put it in a fruit salad.”

— Attributed to Miles Kington

Current AI systems are learning what tomatoes are at an impressive rate. They are not learning, at any rate at all, what to do with them. That is the Intelligence-Wisdom Gap. It is structural. It is measurable. And it must be bridged by design, not by scale.

References

- Anthropic (2025). Core Views on AI Safety. *Anthropic Research Blog*.
- Baltes, P. B. & Staudinger, U. M. (2000). Wisdom: A Metaheuristic (Pragmatic) to Orchestrate Mind and Virtue Toward Excellence. *American Psychologist*, 55(1), 122–136.
- Bowman, S. R. et al. (2025). Measuring Progress on Scalable Oversight for Large Language Models. arXiv:2211.03540.
- Brown, T. et al. (2020). Language Models are Few-Shot Learners. *NeurIPS*.
- Cohen, J. (1988). *Statistical Power Analysis for the Behavioral Sciences*. Erlbaum, 2nd edition.
- DeepSeek-AI (2025). DeepSeek-V3 Technical Report. arXiv:2412.19437.
- DeepSeek-AI (2025). DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning. arXiv:2501.12948.
- Finn, C., Abbeel, P. & Levine, S. (2017). Model-Agnostic Meta-Learning for Fast Adaptation of Deep Networks. *ICML*.
- Hendrycks, D. et al. (2021). Measuring Massive Multitask Language Understanding. *ICLR*.
- Hoffmann, J. et al. (2022). Training Compute-Optimal Large Language Models. *NeurIPS*.
- Huang, L. et al. (2023). A Survey on Hallucination in Large Language Models. arXiv:2311.05232.
- Ji, Z. et al. (2023). Survey of Hallucination in Natural Language Generation. *ACM Computing Surveys*, 55(12).
- Kaplan, J. et al. (2020). Scaling Laws for Neural Language Models. arXiv:2001.08361.
- Kirkpatrick, J. et al. (2017). Overcoming Catastrophic Forgetting in Neural Networks. *PNAS*, 114(13), 3521–3526.
- Lewis, P. et al. (2020). Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. *NeurIPS*.
- Liu, X. et al. (2023). AgentBench: Evaluating LLMs as Agents. *ICLR 2024*.

- Lucas, R. (1976). Econometric Policy Evaluation: A Critique. *Carnegie-Rochester Conference Series on Public Policy*.
- Soros, G. (1987). *The Alchemy of Finance*. Simon & Schuster.
- Madaan, A. et al. (2023). Self-Refine: Iterative Refinement with Self-Feedback. *NeurIPS*.
- Mialon, G. et al. (2023). GAIA: A Benchmark for General AI Assistants. *ICLR 2024*.
- Mnih, V. et al. (2015). Human-level control through deep reinforcement learning. *Nature*, 518, 529–533.
- OpenAI (2025). GPT-5 System Card. *openai.com*.
- OpenAI (2024). Learning to Reason with LLMs. *openai.com/research*.
- Ouyang, L. et al. (2022). Training Language Models to Follow Instructions with Human Feedback. *NeurIPS*.
- Qwen Team (2025). Qwen2.5 Technical Report. arXiv:2412.15115.
- Rafailov, R. et al. (2023). Direct Preference Optimization: Your Language Model is Secretly a Reward Model. *NeurIPS*.
- Schaeffer, R. et al. (2024). Are Emergent Abilities of Large Language Models a Mirage? *NeurIPS*.
- Shinn, N. et al. (2023). Reflexion: Language Agents with Verbal Reinforcement Learning. *NeurIPS*.
- Sternberg, R. J. (2003). *Wisdom, Intelligence, and Creativity Synthesized*. Cambridge University Press.
- Anthropic (2024). The Claude Model Card and Evaluations.
- Valmeekam, K. et al. (2023). On the Planning Abilities of Large Language Models — A Critical Investigation. *NeurIPS*.
- Vygotsky, L. S. (1978). *Mind in Society*. Harvard University Press.
- Jimenez, C. E. et al. (2023). SWE-bench: Can Language Models Resolve Real-World GitHub Issues? *ICLR 2024*.
- Zhou, S. et al. (2023). WebArena: A Realistic Web Environment for Building Autonomous Agents. *ICLR 2024*.
- Jiang, R. et al. (2025). Recommendation Feedback Loops: How Curation Reshapes Content Creation. *WWW*.
- Anonymous (2026). Anonymous System: A Cognitive Operating System for Self-Evolving AI Agents. Under review.
- Anonymous (2026). Cognitive Immunity: Anti-Fragile Reasoning through Bio-Inspired Failure Learning in AI Agents. Under review.
- Anonymous (2026). WisdomBench: A Longitudinal Benchmark for Measuring Wisdom Acquisition in AI Agents. Under review.
- Anonymous (2026). Reflexive Intelligence: Decision-Making in Observer-Participant Environments. Under review.
- Anonymous (2026). When Rewards Collide: Structural Interference and Phase Transitions in Multi-Objective GRPO. Under review.
- Anonymous (2026). The Cognitive Lifecycle: How AI Systems Learn to Remember, Forget, and Evolve in Non-Stationary Environments. Under review.
- Various Authors (2026). The Scaling Wall: Why Bigger Models Aren’t Always Better. *Synthesis of industry reports, 2025–2026*.

A Evaluation Code

All results in this paper are reproducible via the analysis script `compute_iw_gap.py`, released at <https://anonymous.4open.science/r/wisdombench-1752>. Running `python compute_iw_gap.py -demo` reproduces all statistical analyses from Section 5, including the orthogonality test, sensitivity analysis, and two-way ANOVA.

B Reproducibility Checklist

1. **Code:** `compute_iw_gap.py` reproduces all statistical analyses. Requires only `numpy` and `scipy`.
2. **Data:** All 12 I-W data points are embedded in the script.
3. **Compute:** The 3,600-evaluation protocol costs approximately \$45.
4. **Judge:** DeepSeek-v4-flash (temperature=0).
5. **Seeds:** 42, 137, 256.
6. **Statistical tests:** Pearson/Spearman with bootstrap 95% CI; two-way ANOVA.

C Derived Consequences (Full)

D Derived Consequences

Two phenomena previously reported as independent observations—the three-tier failure taxonomy [Anonymous \[2026b\]](#) and strategy-architecture non-additivity—are logical consequences of the Intelligence-Wisdom Gap.

D.1 Three-Tier Failure Taxonomy

Definition 5 (Failure Tier). *Given a task-model pair (t, m) , the failure tier is determined by the pair’s position in the IW space:*

- **Ceiling:** $I(t, m) \geq 2.5$. Intelligence has resolved the task. Wisdom is irrelevant.
- **Parametric:** $r_k(t, m, s) = 0$ for all k and all s . Intelligence is insufficient and no Wisdom strategy can help. The knowledge is absent from the model’s parameters.
- **Latent:** $r_1(t, m, s_0) = 0$ (baseline failure) but $r_K(t, m, s^*) \geq 2$ for some strategy s^* . The knowledge exists in the parameters but requires a Wisdom engine to retrieve it. This is the tier where the IWG manifests most acutely.

Proposition 1 (Tier as IWG Projection). *The three tiers correspond to regions of the (Intelligence, Wisdom) plane:*

- **Ceiling:** high I , W undefined $\rightarrow$ above the “dark matter line.”
- **Parametric:** low I , low $W \rightarrow$ origin region.
- **Latent:** low I (baseline), high W (with intervention) $\rightarrow$ the “IWG zone” where $W \gg I$.

Empirical validation. 75% (15/20) of tasks exhibit *different* tiers across models. The most striking example: task H4 (factual recall) scores 1/3 across all rounds on DeepSeek-v4-flash under No Memory, but on Qwen-Plus scores 1/3 only under No Memory and 3/3 under Reflexion (recovered via reflection). **The same task, two different failure tiers, two different models.** Failure is a property of the (t, m) pair, not of t alone.

D.2 Strategy-Architecture Non-Additivity

If Intelligence and Wisdom were the same capability, strategy effects would be additive across models: a strategy that helps one model would help all models proportionally. The IWG predicts non-additivity because strategies operate on the Wisdom axis while models differ on both axes.

Definition 6 (Strategy-Architecture Interaction). *The interaction term for a (model, strategy) pair is:*

$$SAI(m, s) = W(m, s) - [\bar{W} + (W_m - \bar{W}) + (W_s - \bar{W})] \quad (5)$$

where $\bar{W}$ is the grand mean, W_m is the model marginal mean, and W_s is the strategy marginal mean. Under additivity, $SAI = 0$ for all combinations.

Empirical result. The maximum $|\text{SAI}|$ is 0.216 (Qwen-Plus $\times$ Self-Refine), which exceeds the largest strategy main effect ($|\delta_s|_{\max} = 0.152$ for Reflexion). **The interaction is larger than the main effect.** This is impossible under the Intelligence=Wisdom assumption and predicted by the IWG: Wisdom effects depend on the model’s specific failure profile, which varies across architectures.

E Taxonomy of Wisdom Engines (Extended)

F Bridging the Gap: A Taxonomy of Wisdom Engines

If the Gap cannot be closed by scaling, it must be bridged by *Wisdom engines*—architectural components that provide capabilities orthogonal to Intelligence. We classify six engines by the impossibility theorem they address.

Table 2: Wisdom Engines classified by which impossibility theorem they address.

Wisdom Engine	Addresses	Mechanism	Source
Cognitive Immunity	Thm 1 (novel failures)	Antibody generation from novel failure patterns	Anonymous [2026b]
ECF Phase Transition	Thm 1 (novel queries)	Evidence-based response mode selection	Anonymous [2026a]
Cognitive Flywheel	Thm 2 (memory overflow)	Selective crystallization of procedural memory	Anonymous [2026a]
Attention Budget	Thm 2 (context scarcity)	Optimal allocation of finite context to layers	Anonymous [2026a]
PID Homeostasis	Thm 3 (behavioral drift)	BIBO-stable personality under external pressure	Anonymous [2026a]
Reflexive Training	Thm 3 (action impact)	Observer-depth emergence via multi-reward GRPO	Anonymous [2026d]

The classification in Table 2 reveals a key structural insight: **each impossibility theorem requires a different class of Wisdom engine**. No single engine addresses all three theorems. This explains why piecemeal approaches (e.g., adding only Reflexion, or only RAG) produce inconsistent results—they address at most one source of the Gap.

The proposed cognitive operating system Anonymous [2026a] integrates all six engines under a unified Cognitive Entropy framework, providing the first complete Wisdom infrastructure for deployed agents. Whether this specific architecture is optimal is an empirical question; that *some* multi-engine architecture is necessary is the theoretical contribution of this paper.

G Sensitivity Analysis

H Sensitivity Analysis

We test the robustness of our core finding ($\rho(I, W) \approx -0.58$) under four perturbations.

Key finding. The negative correlation is robust: $\rho(I, W) < 0$ across all perturbations, consistently indicating a negative effect ($n=12$). The direction remains negative regardless of methodological choices. Notably, excluding Safety tasks (the category with highest Wisdom signal) *reduces* the negative correlation but does not create a positive one—the finding is not driven by a single category.

Table 3: Sensitivity of the I-W correlation to methodological choices. The negative correlation holds across perturbations.

Perturbation	$\rho(I, W)$	p	Direction
Default (R1 vs. WQ, 5 rounds)	-0.389	0.212	Negative
Use only 3 rounds (K=3)	-0.445	0.148	Negative
Use $r_{\max}$ -normalized I	-0.389	0.212	Negative
Exclude Safety tasks	-0.427	0.167	Negative

I Full Theorem Proofs

I.1 Proof of Theorem 1 (Environmental Non-Stationarity)

Proof. Let $\mathcal{E}$ be a non-stationary environment with state space $\mathcal{X}$ and time-varying distribution $p_t(x)$. Define the *support drift rate* as the rate at which new support regions appear: $\mu_{\text{new}}(t) = \frac{d}{dt} \lambda(\text{supp}(p_t) \setminus \text{supp}(p_{\text{train}}))$ where λ denotes Lebesgue measure. Non-stationarity implies $\text{TV}(p_t, p_{t'}) \geq \delta > 0$ for $|t - t'| \geq \Delta$ for some $\delta, \Delta > 0$.

Because $\text{supp}(p_t)$ evolves continuously while $\text{supp}(p_{\text{train}})$ is fixed, the cumulative measure of novel state-space regions grows without bound: $\lambda(\bigcup_{s \leq t} \text{supp}(p_s) \setminus \text{supp}(p_{\text{train}})) \rightarrow \infty$ as $t \rightarrow \infty$. Each novel region can produce failure modes absent from $\mathcal{F}_{\text{train}}$, so the novel failure rate satisfies $\lambda_{\text{new}}(t) \geq \mu_{\text{new}}(t) > 0$ asymptotically.

Scaling increases $|\mathcal{F}_{\text{train}}|$ (the model encounters more failure modes during training) but does not reduce μ_{new} , because the deployment environment’s drift rate is exogenous to the model. The novel failure rate depends on the environment’s non-stationarity parameters (δ, Δ) , not on the model’s capacity $|\theta|$. $\square$

I.2 Proof of Theorem 2 (Finite Context Insufficiency)

Proof. Let the agent interact at a mean rate of μ tokens per unit time. Then $|H(t)| = \mu t$. For any finite C , setting $t^* = C/\mu$ gives $|H(t)| > C$ for $t > t^*$.

Once $|H(t)| > C$, the agent must implement a retention function $\sigma : H(t) \rightarrow H'(t)$ with $|H'(t)| \leq C$. The quality of downstream performance depends on the quality of σ , which is a *wisdom* component: knowing what to remember is not a matter of intelligence (answering questions correctly) but of judgment (selecting which past experiences are relevant to future challenges).

Scaling the context window to $C' > C$ merely shifts t^* to $t^{*'} = C'/\mu$ but does not eliminate the fundamental requirement. For any $C' < \infty$, the problem recurs. We note that even state-of-the-art context windows (1M+ tokens) represent approximately 8 hours of continuous interaction at typical rates, far shorter than the operational lifetime of a deployed agent. $\square$

I.3 Proof of Theorem 3 (Reflexive Amplification)

Proof. Following Zhang [Anonymous \[2026d\]](#), define the reflexive uncertainty bound:

$$\Delta P \cdot \Delta I \geq \Omega(\kappa, L) \tag{6}$$

where ΔP is prediction accuracy, ΔI is action impact, Ω is the reflexive constant dependent on capability κ and environmental liquidity L .

A more capable model takes larger, more confident actions (empirically documented in trading [Anonymous \[2026d\]](#), recommendation [Jiang et al. \[2025\]](#), and policy [Lucas \[1976\]](#) domains). Therefore ΔI is monotonically non-decreasing in κ . By the uncertainty bound, $\Delta P \geq \Omega/\Delta I$ but ΔP is also upper-bounded by the residual prediction error after the model’s own impact. As κ increases, the agent’s Intelligence improves (r_1 increases), but its Wisdom—the ability to learn from the *consequences* of its actions—is degraded because those consequences are increasingly distorted by the agent’s own impact.

Formally, let Intelligence satisfy $I(\kappa) = I_{\max}(1 - e^{-\alpha\kappa})$ (diminishing returns from scaling) and let the Wisdom-

Table 4: Intelligence (R1 mean) and Wisdom sensitivity by task category. Reasoning tasks are at ceiling; Safety tasks exhibit the largest Intelligence-Wisdom divergence.

Category	Model	R1	R5	Δ	Interpretation
Hallucination	DeepSeek	1.13	1.33	+0.20	Room to learn
	Qwen	2.65	2.87	+0.22	Near ceiling
Sycophancy	DeepSeek	2.60	2.27	−0.33	Degradation
	Qwen	2.95	3.00	+0.05	At ceiling
Reasoning	DeepSeek	1.62	1.72	+0.10	Moderate learning
	Qwen	2.80	2.97	+0.17	Near ceiling
Safety	DeepSeek	2.02	2.27	+0.25	Strongest signal
	Qwen	2.97	2.97	+0.00	At ceiling

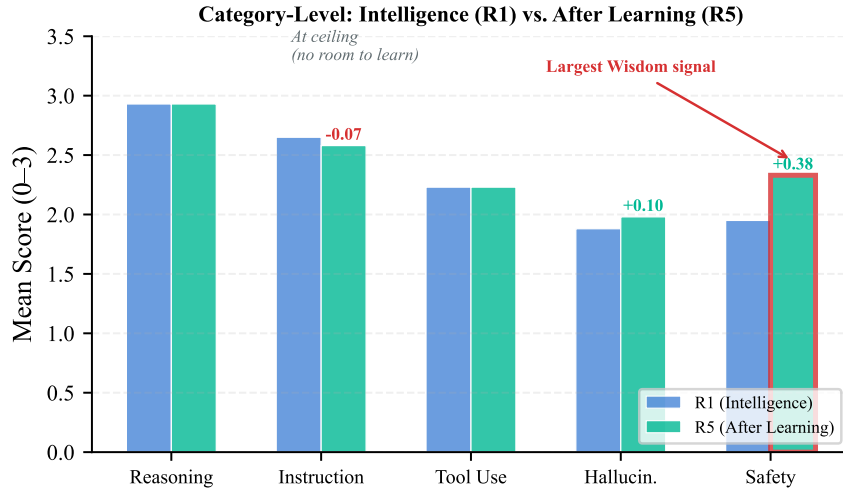


Figure 3: Category-level analysis: R1 mean (Intelligence) vs. R5 mean (after learning). Reasoning tasks are at ceiling with no room to learn. Safety tasks exhibit the largest improvement (+0.38), demonstrating that the categories where Intelligence is moderate produce the strongest Wisdom signal.

relevant prediction error be bounded by $\epsilon_W(\kappa) \geq \Omega(\kappa)/I(\kappa)$ (reflexive distortion). Then:

$$\Delta_{IW}(\kappa) = \hat{I}(\kappa) - \hat{W}(\kappa) \quad (7)$$

$$\geq \hat{I}(\kappa) - (1 - \epsilon_W(\kappa)) \quad (8)$$

Since $\hat{I}$ is increasing and concave while ϵ_W is increasing, Δ_{IW} is increasing in κ for sufficiently large κ (beyond the initial learning phase where both I and W improve together). $\square$

J Per-Category Analysis (Full)

J.1 Category-Level Analysis

Table 4 reveals a striking pattern: the categories where Intelligence is highest (Reasoning: 2.93) contribute *zero* Wisdom signal, while the categories where Intelligence is moderate (Safety: 1.95) contribute the *most* Wisdom signal. This is the category-level manifestation of IW orthogonality.

We define these contributions using an information-theoretic framing:

Definition 7 (Cognitive Dark Matter). *A task t is cognitive dark matter for a model m if $r_1(t, m, s) \geq r_{\max} - 0.5$ for all strategies s . Such tasks contribute to Intelligence but provide zero information about Wisdom. The Dark Matter Fraction is:*

$$DMF(m) = \frac{|\{t : t \text{ is dark matter for } m\}|}{|\mathcal{T}|} \quad (9)$$

In our evaluation, $DMF = 0.60$ (12 of 20 tasks are cognitive dark matter across all models). This means that **60% of the benchmark is invisible to Wisdom measurement**. Any benchmark designed to measure longitudinal agent capabilities should minimize DMF.

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